

# ArchaeoSight Requirements

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ArchaeoSight

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Accepted as baseline requirements for the project:

For the client: \_\_\_\_\_

For the team: \_\_\_\_\_

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# 1. Introduction

There is much about the societies and cultures of the past that we do not know, from who they were to what they valued. Archaeology and anthropology are the unique and diverse fields of study that aim to lessen this gap in our knowledge by recovering and interpreting the material remains left behind by people of the past. Archaeologists' specialties are broad, and they can span from studying the remnants of our ancestors from several million years ago up to the Anthropocene (ca. 12,000-15,000 years ago). Archaeologists do much more than just "dig." Today, archaeologists use a number of modern techniques and tools such as photogrammetry, light detection and ranging (LiDAR), ground penetrating radar (GPR), and portable X-ray fluorescence (pXRF) to guide their recovery efforts. They can also work in a number of other contexts besides conducting fieldwork. In government agencies, they might be responsible for managing and protecting archaeological sites located on public land. Others may work in museums, historical sites, or as administrators who manage programs related to research, collections, and exhibitions. Academic archaeologists might work at colleges and universities to teach students, as well as actively research and publish within their field, such as our own client.

Our client and sponsor, Dr. Kayeleigh Sharp, has been researching the Lambayeque region of northern Peru since 2010 and currently directs the Northern Gallinazo Archaeological Research Project (PIAGN) that began in 2022. Dr. Sharp prioritizes the research of academically overlooked archaeological groups, and her work in Lambayeque aims to reveal the region's past defining social and economic relationships, the complexities of which underlie the phenomenon she calls "economic complementarity." The area of Lambayeque is a significant cultural center for the ancient Andean civilization of South America, being home to one of the most ecologically and culturally diverse locations on the planet. There, Dr. Sharp and her team have been studying numerous archaeological sites, aiming to broaden our society's understanding of Lambayeque life and culture during the first millennium.

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## 2. The Problem

Our client, like many other archaeologists, is equipped with highly specialized devices such as portable X-ray fluorescence spectrometers, referred to as pXRF. These tools are highly effective in non-destructive elemental analysis, capable of the identification of elemental composition onsite. It provides versatile and high-level analysis of soil, architecture, and different artifacts. The basis of ArchaeoSight is centered around combating the basic weaknesses of this tool (in particular, limited spatial applicability). We were able to observe the standard workflow of our client. She gathers her samples and analyzes one sample at a time. The process of this screening consists of pointing the pXRF tool directly at the sample for several seconds or possibly minutes. The elemental composition is then displayed on the screen and prepared for exportation. After the studies are completed and investigations finished, data collected by the pXRF tool is harvested and then transferred to Dr. Sharp's laptop, exported as CSV or PDF files.

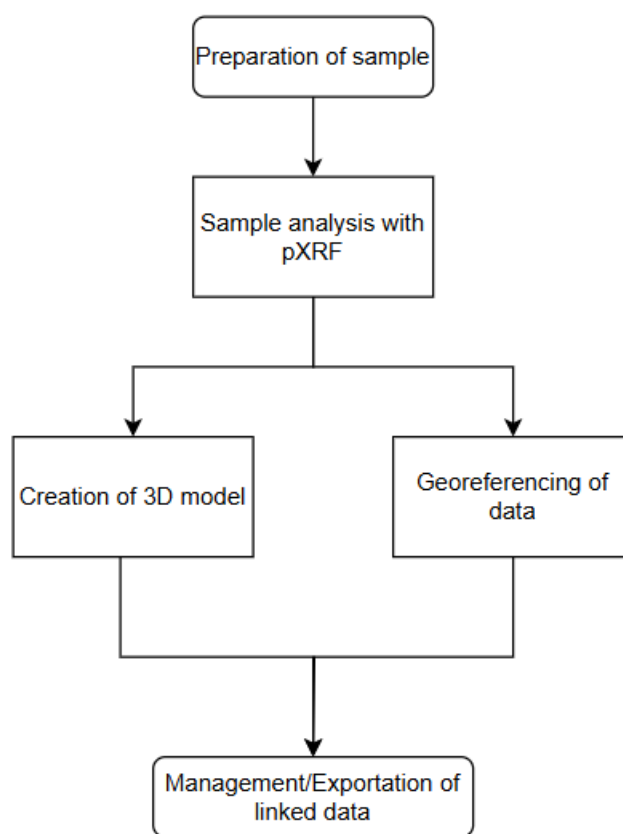
**Figure 1: pXRF SciAps X-200**



While this tool is key to the work of our client, it can prove to be a tedious process due to its spatial limitations. The pXRF tool is primarily used for surface analysis; it is not capable of detection in bulk or anything beneath the surface. It is a highly sensitive tool, as high moisture can affect its readings, making them occasionally misleading. It is important for Dr. Sharp to be able to prepare her samples to provide ideal readings. For the best nondestructive results, direct interaction with the environment is highly recommended and often required. With localized, single-point readings, sampling an entire site is not possible. Our client is looking for specific compounds and metals such as copper and pottery recipes. Having to analyze untargeted samples and georeference them during data post-processing is extremely time consuming and impractical; field data collection with pXRF is far better suited to working from known to unknown rather than spending too much time on searching.

This data collection process can be time-consuming, but it is only half the battle. The management and integration of this data is the remaining challenge. For our client to properly use the data she has collected, it is important to be able to compare those results and test them against their locations and geographic identity. This can often require a few different third-party tools. The processes incorporated by our client include geographic information system (GIS) mapping and 3D imaging for creating interactive models. The usage of these many different platforms can be difficult to keep track of, and interpretation of collected data can add to this dilemma. Some pXRF tools such as the SciAps X-200 pre-process the data it collects from analysis, making the results readable and ready for adaptation. However, many other models do not have this feature, further complicating the integration of data.

**Figure 2: Dr. Sharp's Original General Workflow**



Above is the standard workflow used by Dr. Sharp, her colleagues, and likely many other archaeologists and anthropologists alike. Due to the pXRF's spatial weaknesses, it's very important that a given sample is properly conditioned and prepared for analyzation. When a sample is prepared, pXRF analysis is then performed. Analysis of a single sample typically takes about 90 seconds, analysis of a multitude of samples can take up to hours of time. After sample analysis,

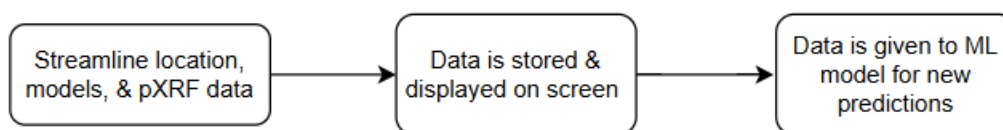
the client must then create 3D models of the sample area's environment, and then georeference that data to a location. Both of these tasks are done with third-party tools that require extensive management to organize and link manually.

Despite its limitations, the pXRF tool remains essential for archaeologists, providing fast elemental analysis in real time that continues to enhance field research. Overcoming its spatial constraints, environmental sensitivity, and data management challenges requires methodological refinements and technological integrations. With the incorporation of GIS mapping and 3D imaging, Dr. Sharp has been able to maximize the efficiency of her research. But the trade-off of this workflow is keeping track of all the tools used and all data found.

### 3. Our Solution

For the optimization of Dr. Sharp's workflow, ArchaeoSight should aim to simplify the flow of data processing and interpretation by streamlining the integration of pXRF results with their geographic locations and respective 3D models all into one mobile application. Having these features all in one place can help her and other archaeologists conduct precise and insightful fieldwork with little hassle. This is all in aid of the standard archeologist's workload, but the true value in ArchaeoSight lies in how we use all of the data that is collected. As seen in the simplified workflow below, our main goal here is to combine pXRF analysis with largescale GIS mapping, link and store that data, and then use ML to make predictions based on this data of other areas that could be rich in anthropogenic activity.

**Figure 3: ArchaeoSight's Workflow**



The predictions made by ArchaeoSight's ML model will be used for guiding our client and other archaeologists alike to potential areas of investigation and different areas of interest. This ML model will not only make predictions but should also be able to provide a level of certainty in its predictions (initially, a minimum of 70% accuracy). By feeding the ML model the collected data, its goal will be to identify the possible distribution of the elemental composition in the area, highlighting hotspots of anthropogenic activity over the 3D modeling. Depending on the results found, the model will learn from its mistakes and successes in order to increase accuracy of future predictions.

ArchaeoSight's convenient data management and the insight gathered from the predictive ML model together will ensure a positive impact on the workflow of our client. The streamlining of data should improve overall efficiency of investigations, allowing for easy management and storage of collected data, and associating that data with GPS location without having to utilize

multiple platforms. The ML model will simplify the process of discovery, as the predictions made can lead to other materials of interest. Given the success of these predictions, providing suggestions of where samples should be taken should shorten the amount of time taken during these investigations and allows the client to gain a better understanding of what could be available in the area, increasing level of insight and decreasing the amount of unnecessary interaction with the environment. ArchaeoSight brings all the essential tools and values of archaeologists together creating a lighter workload to simplify the process altogether, along with a predictive ML model suited for fueling more discovery with its prediction of anthropogenic hotspots.

## 4. Requirements

This section outlines the full set of requirements that define the functionality, performance expectations, and environmental constraints of the ArchaeoSight system. These requirements will serve as the foundation for development and provide a clear, measurable basis for validation project success. Derived through extensive collaboration with our client, Dr. Kayeleigh Sharp, and grounded in the specific needs of archaeological fieldwork, the requirements are structured into three categories:

- **Functional Requirements:** Define the core capabilities the system must support, including data integration, machine learning, collaboration, and user authentication.
- **Performance Requirements:** Specify expectations for system responsiveness, reliability, and usability, particularly on mobile devices used in remote locations.
- **Environmental Requirements:** Describe the technical and operational constraints imposed by the physical field environment, client hardware, and cross-platform deployment needs.

Together, these requirements form the contractual baseline for development and acceptance. They are designed to ensure that ArchaeoSight provides archaeologists with a robust, intuitive, and predictive tool for conducting efficient, data-driven field investigations.

### 4.1 Functional Requirements

#### 4.1.1 pXRF Data Integration

The system shall provide support for integrating and managing portable X-ray fluorescence (pXRF) data, specifically from devices used by the client such as the SciAps X-200 and Bruker series tools. These devices export readings in .csv format, which must be uploaded into the ArchaeoSight app for analysis and visualization.

**Functional Goal:**

- Allow users to select and import .csv files generated by pXRF devices.
- Validate the structure of the CSV file to ensure compatibility with the system's data model.
- Associate each pXRF data entry with specific grid locations on the site model based on user input or GIS data.
- Enable manual correction or review of pXRF readings.

#### **Use Case Example:**

While in the field, a user exports a batch of readings from their SciAps device via Wi-Fi. The user then selects the exported CSV file through ArchaeoSight file picker, loads it into the app, and reviews the data associated with grid cells on a 3D model of the site.

### **4.1.2 User Authentication**

To protect sensitive archaeological data and enable secure collaboration, the application shall support user authentication.

#### **Functional Goals:**

- Provide a login system supporting unique usernames and secure passwords.
- Allow users to register accounts and reset passwords.
- Maintain authenticated sessions locally during offline use and re-sync them when the device is back online.

#### **Use Case Example:**

A field researcher logs into ArchaeoSight using their registered credentials. This allows them to access the data set from their account.

### **4.1.3 Data Syncing**

ArchaeoSight must support seamless synchronization of data between local storage and a remote cloud service to ensure data persistence and collaboration across devices and users.

#### **Functional Goals:**

- All data can be collected and stored locally without requiring internet access.
- Support automatic and manual syncing to a cloud backend when a network connection becomes available.
- Implement differential syncing to upload only new or changed data.
- Handle synchronization conflicts gracefully, prompting users to resolve discrepancies as needed.
- Include visual indicators for sync status and success/failure feedback.

### Use Case Example:

A user collects 20 pXRF readings in the field without internet. Later that day, upon connecting to Wi-Fi at basecamp, ArchaeoSight automatically begins uploading new data to the client's database, showing a progress bar and success confirmation.

## 4.1.4 Machine Learning Model

The meat of this application is the ML model. The model will take multiple kinds of input in two distinct sections. The first section of input is 3D models of an archaeological site.

- These models are generated through an application called Scaniverse. They are exported in either GLB or GLTF formats, which are two formats supported by flutter\_3d\_controller (Flutter's 3D model package).
- The second possible form of input for the first section is 2D images. These can be used as either replacements or (preferably) compliments for the 3D models.
- These will be overhead images of an archaeological site.

The second section will use pXRF readings from the site that the 3D models/images represent. The idea is that the site is split into a giant grid and each pXRF reading represents a cell in the grid (the real-life scale of the grid is about one reading per 50 cm, max).

To predict areas of human activity, we chose a gradient boosting decision tree. The way GBDTs work is they start with an initial guess, then look at why the guess was bad (these are called residuals). Then a second guess is formed between two models instead of one, but this time, both models know what went wrong the first time, so they don't do that again. This process happens repeatedly, until the threshold is met, where the model becomes confident in its answer. This model will be trained on existing data and applied to every cell in the grid. This will allow us to find multiple hot spots, if multiple potentially exist. The reason this algorithm is perfect for what we are doing is because it can find patterns where we can't see them. It can see the correlation between materials and possible human activity. GBDTs also get better at predicting when you have data that is associated to each other. In this case, our 3D grid cells each have a pXRF reading tied to them, which will help the model find patterns. Dr. Sharp has 3D models of places she has previously excavated, along with pXRF data for those sites, which will be the perfect training set. As for what this ML model should output, it will place a heatmap over the model, with marked zones representing a high level of confidence that evidence of human activity should be found there.

## 4.1.5 Accepting Local Files

The ML model will need to take in input in a formatted way, so it can easily give a response. The input consists of a 3D model and pXRF data associated with that model. The 3D models will be in either GLTF or GLB format. Flutter allows you to choose files on your device, so once a

model is made, we can download it, then choose the downloaded file within the app. The pXRF data is a little trickier. To get the pXRF data onto the app, we must do a few things.

- First, gather the pXRF data as normal. Then you need to go to the device's home page and click the “Export” button. This will create a folder called ‘export’.
- Next, connect the pXRF to a mobile hotspot. Then, on another device, type in the IP address of the pXRF device into a web browser (you’ll need to add the port number at the end). Now you’ll see the folders/files for the pXRF device.
- From there you can download them. Then inside the app you can select all of the pXRF files (they come in CSV format) and load them into the app.

## 4.2 Performance Requirements

### 4.2.1 Cross Platform Mobile Application

The first of the performance requirements for our application is that it must be able to run on both Android and iOS devices. Dr. Sharp and her team use a variety of mobile devices out in the field, so we must be able to accommodate all of them. We will do this through a framework called Flutter. Flutter uses the Dart programming language to ensure a seamless experience between Android and iOS devices, achieving at least 98% feature parity across platforms. Dart (with Flutter) is the best option for a few reasons.

- Dart uses ahead-of-time (AOT) compilation to convert directly to machine code, which means it can run directly on the CPU, offering great performance that feels native. This is especially important given our deployment targets:
- Dart is also optimized for UI performance and rendering animations, which will be important for rendering 3D models and generating heatmaps in the field.
- Dart uses a high-performance garbage collector that is incremental and generational, allowing it to clean up memory quickly and predictably without causing noticeable pauses. This helps maintain smooth animations and responsiveness, especially on mid-range devices.

On top of using Flutter as our mobile framework, we have laid out device specific requirements for the application.

- Android devices running API levels 26–34 (Android API level 26 correlates to devices made in 2017–18 onward) and iOS devices running iOS 14.0 or later (iOS 14.0 correlates to iPhones made in 2015 and after).
- Hardware with at least 4GB of RAM.
- Processors equivalent to the Snapdragon 662 or Apple A10.

## 4.2.2 Non-technical/Friendly User Interface

The second performance requirement we have is user-friendly visualization and ease of use. It is important that the archaeologists working with this application are able to easily traverse the application without the need for external assistance. This allows archaeologists of any technical level to use the app confidently and efficiently without wasting time figuring out how it works. The way we will implement this is through the following:

- The interface will include intuitive UI elements such as clearly labeled icons, a consistent navigation bar, and contextual tooltips to guide users through unfamiliar features.
- To support quick onboarding, the app will also feature optional guided walkthroughs and a help section accessible from any screen.
- For visual data like heatmaps, results will be displayed in a simple grid format with color-coded confidence ranges of 0–20%, 21–40%, 41–60%, 61–80%, and 81–100%.
- Map interactions support familiar gestures such as tapping, pinching, and dragging, and all interactive elements will be at least 48 by 48 pixels to accommodate gloved use.
- Spacing between buttons will meet usability standards (minimum 8 pixels) to prevent accidental taps.
- The visual design will also account for field conditions: brightness levels optimized for 400 nits to maintain screen visibility in bright outdoor environments, and text and interface colors will conform to accessibility standards for contrast and readability—including support for high-contrast and large-text modes.
- The interface will maintain sub-500ms response times for 95% of user actions, with no interaction exceeding 1 second without visual progress indicators.
- A multi-threaded architecture ensures smooth performance, with a dedicated UI thread maintaining 60 frames per second during normal use and a minimum of 30 frames per second during heavier tasks.
- Long-running processes such as pXRF data import or model inference will run in the background, with live progress updates every 250ms and user-controlled cancelation options for operations taking longer than 10 seconds.

Together, these usability-focused UI decisions ensure the application remains intuitive, responsive, and reliable for field use, regardless of the user's technical background.

## 4.2.3 Offline-first Workflow

Because this application will be used in archaeological fieldwork, it is important that all core functionalities work without an internet connection. To support the archaeologists' offline workflow, the following must be implemented:

- The app must locally store about a day's worth of data, which is approximately 50 pXRF samples and one to two 3D models and 2D images.
- To ensure users have enough space, the app will be designed to hold at least 150 pXRF samples and up to three 3D or 2D files at once. This allows for multiple days of work without requiring immediate access to a network.
- All data will be stored in an efficient, structured local database or filesystem to ensure quick access and reduce memory footprint.
- The machine learning model used for analysis will be pre-trained and embedded within the application binary, allowing it to run entirely offline without relying on any remote server or internet connection.
- During offline use, users will still be able to perform tasks such as data entry, model-based analysis, and image capture.
- Once internet connectivity is restored, the app will initiate a connection to Dropbox to synchronize the locally collected data. This synchronization will use differential updates—transferring only new or modified files—to minimize bandwidth usage.
- Transfers will be resumable to handle intermittent connectivity, and compression will be applied where possible to optimize performance.

This hybrid model ensures that archaeologists can work efficiently in the field while preserving data integrity and syncing to the cloud as soon as a connection becomes available.

#### 4.2.4 Machine Learning Model Requirements

For the machine learning part of this application, it is important that the model runs quickly and gives accurate predictions. The GBDT approach proves ideal for archaeological feature detection as it excels at handling the mixed categorical and continuous data from pXRF readings while offering natural interpretability of feature importance—allowing archaeologists to understand which elemental markers most strongly indicate human activity. Model optimization includes:

- 8-bit quantization reducing model size by ~75% with accuracy loss under 1.5% compared to the full-precision model.
- By implementing core algorithm components in C++ rather than Dart, the system leverages low-level memory management and pointer arithmetic to achieve 2.5-3x faster execution compared to high-level language implementations.
- This C++ implementation is particularly beneficial for GBDT models, as their tree-traversal operations benefit significantly from cache-friendly memory layouts and branch prediction optimizations available at the lower level.
- The model's C++ core, wrapped with a FFI (Foreign Function Interface) layer, maintains Flutter's cross-platform benefits while achieving superior performance for processing the complex soil composition data.

- The pre-training approach, where model weights are optimized on high-performance computing infrastructure before deployment, eliminates resource-intensive on-device training while preserving the model's ability to detect subtle elemental patterns indicative of archaeological significance.

For a full run through, the model should take in a site ground map, a 3D model/2D overhead images of the site, and pXRF data for those images and models, and return a response in the form of a heatmap in no longer than 15 minutes. Dr. Sharp expressed the fact that archaeological methods take time, so the model doesn't need to give a response in seconds. Since the model has ample time to run, it should allow for confident and complete responses. Given this large time constraint, the model should give responses with at least 70% accuracy.

## 4.3 Environmental Requirements

ArchaeoSight is being designed for use in field conditions where internet access may be intermittent, hardware may be resource-constrained, and compatibility with specialized scientific instruments is essential. The environmental requirements reflect the operational constraints imposed by the archaeological fieldwork context.

### 4.3.1 Platform Compatibility

The application must run on a variety of modern mobile operating systems to support the diverse devices used by archaeologists.

- **Requirement:** The system will be compatible with Android (API level 26+) and iOS (version 14.0+).
- **Justification:** Field researchers use a mix of Android tablets, iPhones, and laptops. Ensuring cross-platform compatibility maximizes accessibility for all team members.

### 4.3.2 Internet Connectivity Constraints

Since excavation sites are often located in remote areas, the application must be functional without internet access while still supporting full synchronization when connectivity is available.

- **Requirement:** The system will operate fully offline, with online features (such as cloud sync and account authentication) enabled when a network becomes available.
- **Justification:** Many archaeological sites lack a reliable internet. Offline-first operation ensures data can be collected, reviewed, and analyzed without interruptions of connectivity.

### 4.3.3 Device Compatibility with pXRF Hardware

ArchaeoSight must work seamlessly with pXRF instruments used by the client. These tools output chemical composition data in specific formats, which the app must be able to parse and interpret.

- **Requirement:** The application will support importing data exported from SciAps X-200, Vanta, and Bruker pXRF devices, specifically in .csv format.
- **Justification:** Dr. Sharp's team uses these pXRF tools in the field. Compatibility with their output formats is necessary for successful integration and analysis.

#### 4.3.4 Hardware Limitations

Given the limited computational power of field equipment, the application must be optimized to function on devices with modest specifications without impacting the user experience.

- **Requirement:** The system will operate on devices with 4 GB of RAM or more without degradation of core functionality or significant slowdown in interface responsiveness.
- **Justification:** Most modern handheld devices operate on 4 GB of RAM or more. Setting this as the baseline ensures responsive performance on these devices and is critical for usability in the field.

### 5. Potential Risks

#### 5.1 Breach of Confidentiality

With the nature of this application, there are some risks that need to be addressed. The first of which is a breach of confidentiality. A breach of confidentiality is when unauthorized users can view data. Even though the data is not sensitive, like SSNs or passwords, it is still our data. Archaeologists worked very hard to collect it, and they will most likely use this data to publish their research. If other parties can view it, they could steal it and use it to create/train their own models or publish findings prematurely.

- **Severity:** High (Could undermine research credibility and lead to intellectual property theft)
- **Effect:** Loss of competitive advantages, potential misuse of data, reputational damage
- **Likelihood:** Medium (Depends on security measures in place, but not highly targeted like financial data)

#### 5.2 Breach of Availability

Another risk is a breach in availability. This is when, for some reason, users are unable to use a service. In the context of our application, this could occur when multiple people are trying

to sync their data to the database. To avoid this, we will have to implement ways to make sure that people are not disconnected from DropBox when trying to sync.

- Severity: Medium (Disrupts workflow but does not permanently damage data)
- Effect: Delays in data collection and analysis, frustration among researchers
- Likelihood: Medium to High (Possible during peak usage if synchronization isn't optimized)

### 5.3 Waste of User's Time

The third, and most important, risk of our app is wasting time. The whole point of this app is to streamline Dr. Sharp and other archaeologists' workflow and predict areas of potential archaeological discovery. If the app doesn't work properly, then the workflow is slowed down, and things don't get done. If the model is not accurate, archaeologists will essentially be guessing where to look, instead of having the ML model guide them to the best spots. If the app fails to do these things, then everyone's time will be wasted and nothing will change.

- Severity: High (Directly undermines the app's core purpose)
- Effect: Reduced efficiency, loss of trust in the tool, potential abandonment by users
- Likelihood: Medium (Depends on model accuracy and app stability, but critical to avoid)

## 6. Project Plan

The development of ArchaeoSight will follow an iterative and milestone-driven approach that prioritizes early delivery of core features, ongoing integration with field data, and frequent client feedback. Given the diverse nature of functional requirements, from machine learning to UI to hardware integration, our timeline is organized around progressive implementation and testing of each major subsystem. Below is an overview of key development milestones.

### 6.1 Project Milestones

| Milestone                   | Description                                                                                             | Target Completion |
|-----------------------------|---------------------------------------------------------------------------------------------------------|-------------------|
| M1. Framework Setup         | Set up Flutter project repository, CI/CD pipeline, and base architecture for cross-platform deployment. | Week 1            |
| M2. PXRF Data Import Module | Implement file picker, CSV parser, and grid association system for pXRF readings.                       | Week 3            |

|                                            |                                                                                                                |         |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------|
| M3. User Authentication                    | Create user registration, login, and session persistence system.                                               | Week 4  |
| M4. 3D Model & Image Import                | Enable upload and preview of GLB/GLTF files and 2D site images.                                                | Week 5  |
| M5. ML Pipeline Integration                | Integrate pre-trained GBDT model for heatmap generation and overlay on site data.                              | Week 7  |
| M6. GIS Mapping Integration                | Enable GPS tagging of samples and spatial visualization through flutter_map or google_maps_flutter             | Week 8  |
| M7. Offline Data Syncing System            | Build local storage, implement differential syncing with Dropbox/Firebase, and test offline-first capabilities | Week 9  |
| M8. UI/UX Refinement and Usability Testing | Conduct testing sessions with archaeologists and improve interface based on feedback                           | Week 10 |
| M9. Final System Testing & Bug Fixes       | Full system integration test, performance tuning, and edge case handling.                                      | Week 11 |
| M10. Client Review & Final Presentation    | Deliver stable release version, collect final feedback, and present final capstone demo.                       | Week 12 |

## 6.2 Development Strategy

Each milestone will follow a development cycle involving:

- Design & Planning: Determine system requirements and select tools/libraries.
- Implementation: Code and test functional components in isolation.
- Integration: Merge with existing modules and resolve compatibility issues.
- Validation: Test functionality with real or mock data and review with the client when applicable.

To stay agile and collaborative, weekly meetings will be held to review progress, assign tasks, and adjust deadlines. GitHub will be utilized for version control and issue tracking, and each milestone will have a designated lead responsible for its timely completion.

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## 7. Conclusion

In modern archaeology, LiDAR, GPR, and pXRF are just some examples of the non-destructive and precise tools archaeologists use to achieve their goals. The problem comes with the tedium of manual data recording due to these tools, especially pXRF, not interfacing conveniently with mobile devices and computers.

ArchaeoSight intends to remedy this by streamlining the pXRF data collection process and allowing the specialized instruments to send data directly to the Android and iOS devices often used by archaeologists in the field. Not only that, but ArchaeoSight will combine the pXRF analyses with location data and 3D mapped regions to use ML to make predictions of areas that are likely to be significant hubs for anthropogenic activity. Our app will also be developed with the consideration of what resources archaeologists have or don't have access to in the field, namely stable internet, ensuring a seamless experience regardless of environmental conditions.

The completion of this document allows us to move forward with the development of ArchaeoSight, ensuring that we and our client are all clear on the scope and goals of this project. At its core, our app will accept pXRF data, guide users to potential sites of interest, and package it all in a responsive app. With our requirements laid out, we have inspired greater confidence within our team and client and may proceed to planning a schedule and prototyping. ArchaeoSight is on track, and we look forward to the possibilities in the field of archaeology and anthropology we will create in the future.

## 8. Revisions

| Version | Revision Description                                                         |
|---------|------------------------------------------------------------------------------|
| 0.1     | Initial Draft of Requirements Document for review by Team Mentor and Client. |